Subject Reference Facts and Worked Examples

Concise facts and worked examples that can support lesson explanations and checks.

Reference material for lesson planning and course-content development.

Contents

1. Computing quick reference

Python range(start, stop, step) includes the start value and excludes the stop value. If only one argument is given, the start defaults to zero. A positive step moves upward; a negative step can move downward when the start and stop values are appropriate.

A Python function can return a value to its caller. print() writes output but does not by itself provide a reusable return value. Lists are ordered collections and use zero-based indexes.

In Git, .gitignore prevents matching untracked files from being added in normal workflows, but it does not erase a file from earlier commits. Merge conflicts require a person to choose the final content.

A SQL JOIN condition determines which rows are connected. GROUP BY creates groups for aggregate calculations. FastAPI uses Python type information and Pydantic models to validate many request fields automatically.

Term	Short reference
for-loop	Iterates over values from an iterable
while-loop	Repeats while a condition is true
JOIN	Combines related rows
GROUP BY	Forms groups for aggregates
404	Requested resource not found
422	Request validation problem in common FastAPI cases

2. Mathematics quick reference

Linear equations preserve equality when the same operation is applied to both sides. Slope is change in y divided by change in x. In $y = mx + b$, m is slope and b is the y-intercept.

The mean is the sum of values divided by the number of values. The median is the middle ordered value. The range is maximum minus minimum. Outliers can affect the mean strongly.

Percentage change compares the difference with the original value. A change from 40 percent to 50 percent is 10 percentage points and a 25 percent relative increase.

Quadratic expressions contain a squared variable term. Factored form can reveal zeros when a product equals zero.

3. Physics quick reference

Average speed equals total distance divided by total time. Velocity includes direction. Acceleration is the rate of change of velocity.

Newton's second law relates net force, mass, and acceleration. Newton's third-law pairs act on different objects. Zero net force is consistent with rest or constant velocity.

Kinetic energy depends on mass and the square of speed. Gravitational potential energy near Earth's surface depends on mass, gravitational field strength, and height.

Units matter. A speed result should include a distance-per-time unit. Force is measured in newtons, energy in joules, and power in watts.

4. Biology quick reference

The cell membrane controls movement of substances. The nucleus contains most genetic material in eukaryotic cells. Ribosomes make proteins. Mitochondria are involved in cellular respiration.

DNA stores genetic information. Transcription produces RNA from a DNA template. Translation uses mRNA information to build a polypeptide.

Photosynthesis captures light energy and stores it in chemical form. Cellular respiration transfers energy from organic molecules into ATP.

Plant cells typically have a cellulose cell wall and may contain chloroplasts. Animal cells do not have cellulose cell walls.

5. Chemistry quick reference

Atomic number equals proton count. Isotopes have the same proton count but different neutron counts. Ions differ in electron count relative to proton count.

Chemical equations are balanced by changing coefficients, not subscripts. Changing a subscript changes the substance.

At room temperature, pH below 7 is acidic, 7 is neutral, and above 7 is basic for dilute aqueous solutions. The pH scale is logarithmic.

Indicators change color over ranges of pH. Neutralization reactions depend on the specific acid and base involved.

6. Business communication quick reference

A professional email should make its purpose clear early, provide enough context, and end with a specific next step or request. A useful subject line helps the reader understand the task before opening the message.

A project update should separate progress, current work, blockers, and next steps. A blocker should state what is blocked and what is needed.

Meeting notes should capture decisions, owners, deadlines, and open questions. A transcript is a raw record; notes select the information that matters after the meeting.

7. Worked Python examples

Loop over a range

```
for i in range(1, 6):
print(i)
```

This prints 1 through 5 because the stop value 6 is excluded.

Running total

```
total = 0
for price in [5, 8, 3]:
total = total + price
print(total)
```

The values of total are 0, 5, 13, and 16.

Function return

```
def double(number):
return number * 2

result = double(4)
```

The returned value is 8 and is stored in result.

8. Worked SQL and API examples

Grouped SQL

```
SELECT users.name, SUM(orders.total)
FROM users
JOIN orders ON users.id = orders.user_id
GROUP BY users.id;
```

Each result row represents one matched user and the total value of that user's orders.

FastAPI request model

```
class ChatRequest(BaseModel):
session_id: str
message: str
```

A request missing message does not satisfy the declared model.

REST example

GET /lessons/12 requests an existing lesson. POST /lessons sends data to create a new lesson. The exact response shape should be documented and consistent.

9. Worked mathematics examples

Two-step equation

For $3x + 5 = 20$, subtract 5 from both sides to get $3x = 15$, then divide both sides by 3 to get $x = 5$. Substitution gives $3(5) + 5 = 20$.

Slope

Between (2, 3) and (6, 11), the change in y is 8 and the change in x is 4, so the slope is 2.

Outlier

For 4, 5, 5, 6, 27, the median is 5 while the mean is 9.4. The large final value pulls the mean upward.

10. Worked science examples

Average speed

A cyclist travels 12 km in 0.5 hours. Average speed is 24 km/h.

Force and acceleration

If the same net force acts on two objects and one has twice the mass, the larger-mass object has half the acceleration in the simplified model.

Balanced equation

$2H_2 + O_2 \rightarrow 2H_2O$ conserves four hydrogen atoms and two oxygen atoms on each side.

11. Worked communication examples

Weak request

Please send this soon.

Clearer request

Please send the revised client report by Thursday at 3 PM so we can review it before Friday's meeting.

Project update

Completed: API authentication and request validation. Current: testing the chat endpoint with separate session IDs. Blocker: waiting for access to the final reference files. Next: finish persistence and prepare demo cases.